KAI FRAZIER

kairfrazier@gmail.com (425) 970-0000 Seattle, WA United States Citizen

aviationnerd12.github.io/Website

WORK EXPERIENCE

Student Front Desk Assistant 07/2026 – Present
UW *Department of Electrical & Computer Engineering*
- Manage intake, triage, and distribution of laboratory shipments, freight, and technical departmental hardware across engineering facilities.
- Administer enterprise physical security systems, managing keycard access provisioning for laboratory facilities and conference rooms.

Advising Navigator 09/2024 – 09/2025
Edmonds College Lynnwood, WA
- Managed front desk operations, multi-channel communications, and appointment scheduling for the academic advising office.
- Triaged student needs and directed them to appropriate academic and financial services.

Technical & Operational Support Intern 08/2023 – 10/2024
Bureau of Alcohol, Tobacco, Firearms and Explosives (ATF)
- Provisioned and configured mobile communication hardware for agency personnel while maintaining compliance with federal security protocols.
- Handled secure data processing and document chain-of-custody protocols for sensitive digital media and agency records.

Pharmacy Assistant 08/2023 – 04/2024
CVS *Pharmacy* Lynnwood, WA
- Executed prescription fulfillment, managed inventory, and processed insurance claims in a high-volume retail environment.
- Maintained strict adherence to federal health regulations and HIPAA protocols.

EDUCATION

Bachelor of Science in Electrical and Computer Engineering 09/2025 – 03/2028
University of Washington Seattle, WA
GPA 2.98/4.0
Relevant Coursework: Semiconductor Devices (EE 331), Circuit Theory (EE 233), Digital Circuits & Systems (EE 271), Computer Hardware Skills (EE 201), Signal Processing (EE 241), Advanced Technical Communication (EE 393)

Associates of Science 09/2024 – 08/2025
Edmonds College Lynnwood, WA
GPA 3.48/4.0

TECHNICAL SKILLS

Engineering Software & Design
Ansys HFSS, LTspice, PCB Layout, Microstrip Synthesis, RF Simulation

Bench Equipment & Testing
Digital Storage Oscilloscopes, Digital Multimeters (DMM), Function Generators, DC Power Supplies, Soldering & Rework

ENGINEERING PROJECTS

1090 MHz Parallel-Coupled Microstrip Bandpass Filter
Tools: Ansys HFSS, SMath, Python, Fortran
- Designed and synthesized a 1090 MHz coupled microstrip bandpass filter for an ADS-B aerospace receiver using Chebyshev prototype low-pass parameters and J-inverters.
- Modeled 3D electromagnetic structures and substrate geometries in Ansys Electronics Desktop (HFSS) to validate S-parameters (S11, S21) and passband transmission.
- Folded half-wavelength resonators into U-shaped hairpins, reducing required board area by over 50% while achieving a 45 MHz passband and a return loss under -15 dB.

Adjustable AC-to-DC Linear Power Supply
Tools: Hardware Design, Linear Regulators, Lab Prototyping
- Designed and built a transformer-fed, full-wave rectified AC-to-DC power converter converting 120V AC line voltage into a variable, low-noise DC output.
- Integrated smoothing capacitors and a potentiometer-controlled Zener shunt network to provide stable voltage regulation across variable resistive loads.

- Measured voltage regulation, stability, and transient behavior using digital multimeters, DC power supplies, and bench oscilloscopes to verify an output range of 9.25 V to 18.65 V.

3-Channel Analog Audio Mixing Console & Multi-Band Equalizer
LTspice, Active Filters, Op-Amps, Keysight DSO

- Engineered an analog audio processing console handling two line-level inputs and one electret condenser microphone channel utilizing an active low-noise preamplifier.
- Designed a reconfigurable 2nd-order active filter topology using LM348N op-amps to create variable band-pass, all-pass, and band-stop (notch) equalizer stages.
- Validated ±15V dual-rail power delivery and measured Bode plot gain responses ranging from +9 dB to -7.9 dB across 250 Hz, 1 kHz, and 4 kHz center frequencies.

Autonomous Mobile Robot & Custom Mixed-Signal Sensor PCB
Tools: KiCad EDA, Schematic Capture, PCB Routing, Hardware Bring-Up, Arduino

- Executed end-to-end schematic capture, custom footprint creation, and 2-layer PCB layout in KiCad for a 7-channel optoelectronic sensor array.
- Routed differential analog trace pairs and integrated emitter-biasing networks to maximize signal integrity and calibrated dynamic range.
- Implemented isolated power planes and decoupling capacitors to protect digital logic lines from motor inductive transients and ground bounce.
- Assembled, soldered through-hole/SMD components, and validated board-level power rails and signal paths using DMMs and bench oscilloscopes.